Lec-1
21/04/26 6:22 PM

[l, r] l r

[a1 a2 a3 -- an]

1 ≤ l ≤ r ≤ n

B. The Fibonacci Segment

time limit per test: 1 second
memory limit per test: 256 megabytes

You have array $a_1, a_2, ..., a_n$. Segment $[l, r]$ $(1 \le l \le r \le n)$ is good if $a_i = a_{i-1} + a_{i-2}$, for all i $(l + 2 \le i \le r)$.

Let's define $len([l, r]) = r - l + 1$, $len([l, r])$ is the length of the segment $[l, r]$. Segment $[l_1, r_1]$, is longer than segment $[l_2, r_2]$, if $len([l_1, r_1]) > len([l_2, r_2])$.

Your task is to find a good segment of the maximum length in array a. Note that a segment of length 1 or 2 is always good.

Input

The first line contains a single integer n $(1 \le n \le 10^5)$ — the number of elements in the array. The second line contains integers: $a_1, a_2, ..., a_n$ $(0 \le a_i \le 10^9)$.

Output

Print the length of the longest good segment in array a.

[a1 a2 a3 a4 a5 a6 a7 a8 a9]

4 6

[a1 a2 a3]

NF

$(10^5)^2 = 10^{10}$

1-1 → a1
1-2 a1 a2
1-3 a1 a2 a3

2-2 a2
2-3 a2 a3

3-3 [a3]

10
1 2 3 5 8 13 21 34 55 89

[1 2 3 5 16 18 34 52 86 0 1 1 2]

4 5 4

1 2 3 5 ×

ans = 84

curr-seq = 2
4

[1 0 0 0 0] ans = 4

[1 2 3 5 16 18 34 52 86 0 1 1 2]
0 1 2 3 4 5 6 7 8 9 10 11 12

for(i=2 i<n i++){
if (a[i-1]==a[i-2]+a[i-3]){
curr-seq++
}
else
curr-seq = 2
ans = max(curr-seq, ans)
}

i=2 → a2 = a1 + a0
i=3 → a3 = a2 + a1
i=4 → a4 = a3 + a2
i=5 → a5 = a4 + a3
i=6 a6 = a5 + a4
i=7 a7 = a6 + a5

Saitama accidentally destroyed a hotel again. To repay the hotel company, Genos has volunteered to operate an elevator in one of its other hotels. The elevator is special — it starts on the top floor, can only move down, and has infinite capacity. Floors are numbered from 0 to s and elevator initially starts on floor s at time 0.

The elevator takes exactly 1 second to move down exactly 1 floor and negligible time to pick up passengers. Genos is given a list detailing when and on which floor passengers arrive. Please determine how long in seconds it will take Genos to bring all passengers to floor 0.

Input

The first line of input contains two integers n and s $(1 \le n \le 100,\ 1 \le s \le 1000)$ — the number of passengers and the number of the top floor respectively.

The next n lines each contain two space-separated integers f_i and t_i $(1 \le f_i \le s,\ 1 \le t_i \le 1000)$ — the floor and the time of arrival in seconds for the passenger number i.

Output

Print a single integer — the minimum amount of time in seconds needed to bring all the passengers to floor 0.

Examples

input
```
2 7
2 1
3 8
5 2
```
output
```
11
```

input
```
5 10
2 77
3 33
8 21
9 12
10 64
```
output
```
79
```

7+1 = 2+1+1 = 4

4+4 = 8+1

9 +=0

11

t = 0+64 = 64+1+1+1+1+1
= 69+1+1
= 71
= 77 77+1
= 78

6 10
2 77
8 33
8 21
9 12
10 64
288

curr = 64+1
= (65, 12) 65
= 66 (21) = 66+1
= 67

[1, 2, 3]

O(n)

log(n)

Example 1:

```
Input: nums1 = [1,3], nums2 = [2]
Output: 2.00000
Explanation: merged array = [1,2,3] and median is 2.
```

Example 2:

```
Input: nums1 = [1,2], nums2 = [3,4]
Output: 2.50000
Explanation: merged array = [1,2,3,4] and median is (2 + 3) / 2 = 2.5.
```

[1 2 3 4]

0 1 2 3 4 5
[2, 3, 4, 11, 18, 16]
[1 3 5 7 9 12 15 17]
1 2 3 4 5 6 7 8

8+6 = 14/2

(8+6)/2 = 3

(8+6)/2 - 3 = 4

max(l1, l2) + min(l1, l2)

2 3 4 11 | 1 8 16
1 3 5 7 | 9 12 15 17

0 1 2 3
2 3 4 16
1 5 7 8 9 10
0 1 2 3 4 5

3 4

23 | 416
157 | 8 910

0+4/2 = 2

6+4/2 - 2 = 3

6+4/2 - 3 = 2

(6+4)/2 - 3 = 2

(m+n/2) - 1

(6+4/2) - 3 = 2

2 3 4 16
1 5 7 8 9

4-4

2-4 = 0

8-4 = 2

0+4/2 = 2

6+4/2 = 2

6+4/2 - 4 = 1

2 3 4 4
1 5 7 8 9 10

4+2/2 + 1 = 2

0+2/2 = 1 0-0/2 = 0

57 88
1 2 34

l1 | l2
57 | 88
12 | 34
- | 57 58
1 | 34 | 4

23 | 44
157 | 8 910
4
234
15 | 7 8 9 10

234 4
1 | 5 7 8 9 10

-2

l1 > l2

0+4/2 = 2

8+4/2 - 2

4+2 = 2

2 3 5 7 9
1 4 6 8

5 | 4
1 4 | 6 8
2 3 5 | 7 9

(5+4+1/2) - 2 =